

ME3043 Production and Operations Management
Semester 05

Final Report

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Executive Summary

Rajarata Coco Lanka is a coconut-based production company that processes husks into coco peat, chips, and fiber. Current operations are efficient, but challenges such as limited husk supply, inconsistent quality control, outdated machinery, and production halts due to weather require optimization. The company faces high labor and transportation costs, limited production capacity, and significant wastage.

Several solutions are proposed. Implementing a demand forecasting system based on historical data will improve predictions of raw material needs, reducing waste and transportation costs. Upgrading machinery, including husk cutting and drying machines, will boost production speed and lower labor costs. Automated drying methods will replace sun-drying, ensuring continuous production regardless of weather. Lastly, optimizing the supply chain will enhance raw material collection and inventory management, improving the overall production process.

Simulations and analyses show that optimizing the production process can significantly increase efficiency, reduce costs, and minimize wastage, ultimately boosting the company's capacity to meet higher market demands. Long-term recommendations include continuous machinery upgrades, quality control enhancements, and regular use of simulation tools for process optimization.

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1 Introduction

1.1 Background

1.1.1 About

Rajarata Coco Lanka follows a systematic and efficient production process to deliver high-quality coconut-based products. The process begins with the collection of suitable coconut husks, which are then separated into different parts. These parts are processed into coco chips, coco peat, and coco fiber by cutting them to the required sizes. Depending on customer requirements, the products can be either washed to remove impurities or left unwashed. The next crucial step is drying, where the moisture content is reduced and maintained below 20% to ensure product quality and longevity. Finally, the dried products are carefully packed, ensuring they are protected during transportation and reach customers in optimal condition. This well-organized production line allows Rajarata Coco Lanka to consistently supply superior products, meeting the diverse needs of the agricultural sector.

1.1.2 Current work conditions

1. Layout

- Open layout with a total space capacity of **half an acre**.
- **60 perches** of space used for **dry coco peat** and **coco chips**.
- The remaining space is allocated for **coco fiber production** and **storing and packaging coconut husks**.
- Additional space is used for **cutting husks** for coco peat production.
- 2 cutting machines for husk cutting, coco-fiber producing machine, coco-peat producing machine under different sizes are the machines used.

2. Labor

- Around 8 **skilled workers**, each trained for **6 months** and 7 unskilled workers trained with senior members.
- Workers are experienced in selecting quality coconut husks without scientific procedures, experienced in all tasks

3. Materials and Required Resources

- Main raw material: **Coconut husks**, sourced mainly from **Anuradhapura** and nearby coconut plantations.
- Husks are collected via 3 **lorries daily according to the maximum capacity** and stored accordingly for production.

- **Machine maintenance** and parts inventory are critical for continuous operation.
- A water storage tank of a capacity 12m³ is used to remove inappropriate acids on coco peels and kept for 12 hours in water for a better stagnation process

4. Production

- **Three main products:** coco peat, coco chips, and coco fiber.
- **Just-in-time (JIT) production methodology** is employed, meaning production starts when raw materials are gathered and scheduled based on demand.
- Production process:
 - Husks are cut, stored in tanks for 12 hours, and dried for **8 hours** using sunlight in the morning.
 - Products are packaged according to consumer requirements.
- Production levels fluctuate throughout the year, influenced by **weather conditions** and **demand**.

5. Scheduling

- Production is scheduled based on **demand forecasts** and **raw material availability**.
- Scheduling adapts to changes in weather conditions, especially during the rainy season when sun-drying isn't possible.
- Demand forecasts are based on **historical data of 3 years** to anticipate supply-demand fluctuations.

6. Inventory Management

- **Raw material inventory** (coconut husks) is managed to minimize transportation and purchasing costs.
- Inventory for **machine maintenance** is essential to avoid downtime.
- Inventory management is influenced by **weather conditions**: production stops during wet weather.
- Regular tracking of **labor, packaging, and distribution costs** to control expenses.

7. Demand Management

- **Demand forecasts** are based on **3 years of historical data** on supply-demand fluctuations under various conditions.
- Production planning is closely linked to demand, with variations considered throughout the year.
- Adjustments are made to show **seasonal fluctuations** in both demand and the availability of raw materials.

1.2 Problem definition

1.2.1 Limitations

1. *Husk collecting problems*

Coconut husks are currently collected only regionally and from the nearby villages around the company only. Therefore, the husk stock per each is less and to fill up the required demand high number of husk sellers required, and it limits the production and inventory cost high as high transportation and labour cost is required. Basically, the supply for the demand gets limited due to this problem and reduce the number of buyers of the product and minimize the profit and maximize cost.

2. *Measuring the quality of husks before production*

The quality of the husks collected are measured only by visual identification only. It's not accurate and in production wastage is high.

3. *Inefficiency on husk cutting machines and inconsistency*

There are two machines used for husk cutting and obtain coco peels according to industry demand. However, the blades used in these machines must be replace according to the coco peel sizes. Therefore, these machines aren't consistent for obtain coco peels for a range of coco peel sizes. Also, these machines required high electricity power and labour cost is high as husks must manually inserted and only one husk can be inserted at a time.

4. *Following the production process in rainy days and husk shortage periods*

The production is limited only for the sunny days as sunset is the main energy source for control the moisture content of coco peels. If there is a deficit in husks during this period there will be a huge wastage and shortage cost. Also, there is a time duration where that the availability of supply becomes less than demand due to the shortage of husks.



Figure 1: Suitable husk selections



Figure 2: Husk cutting



Figure 3: Drying coco chips

1.2.2 Solutions

- **Implement a demand forecasting system** using past data to predict future husk requirements more accurately. This system can use past trends and seasonal patterns to improve buying strategies.
- **Implement more efficient and versatile machinery** Introduce advanced husk cutting machines that can handle a range of coco peel sizes without the need for frequent blade changes, thus improving process flexibility. These machines should feature automated

husk feeding systems to reduce labor costs and increase throughput. Additionally, selecting machines with optimized energy consumption can lower operational costs related to high electricity usage.

- **Waste reduction strategies** Incorporate technology that minimizes husk wastage by ensuring precise cuts and better material utilization, reducing the scrap rate in the production process.
- **Implement alternative drying methods**
- **Improve supply chain management**
- **Production planning and scheduling**

1.3 Project objectives and scope

1.3.1 Aim and Objectives of the project

The aim of the project is to understand the problems in the production line of the company we chose and solve the problems in that production line by introducing production and operations management concepts and further optimizing the processes for better performance.

Objectives

- Develop strategies on supply chain management
- Plan the production and come up with an optimized plan on the production line
- Develop a more flexible production layout
- Utilize capacity for any condition (ex-: weather)

1.3.2 Project scope

- Supply chain management
- Production planning and scheduling
- Process and layout optimization
- Quality control and waste management
- Inventory and resource management
- Cost reduction and profit optimization

2 Methodology

- *Implementation of a demand forecasting system*
- *Implement more efficient and versatile machinery*
- *Implement alternative drying methods*
- *Plan the production process and schedule the production line processes*

2.1 Data collection

Number of husks collected per day – Based on the demand the maximum can be 5000 only

Monthly output – 30tons of coco chips

Main products – Coco chips, coco peal, coco fibre

Operated according to high demand product by time (Currently coco chips)

Stagnated tank capacity – 12m³

Time duration in tank – 12 hours

Time duration used to dry coco chips – 8hours

Area used to dry chips – 60 perches

Number of labours – 15

Currently used motors power – 1HP to 3HP

The data was obtained by the owner and coworkers of the company and from personal inspections for the visits on the company.

2.2 Proposed modifications

- *A new demand forecasting system for manage the supply chain of coconut husks*

The past data of previous months is gathered and identify the demand variation to optimize the supply chain and reduce wastages and minimize costs for transportation, labours and storing.

Also, this system is utilized to be used under a very high demand unless the current demand of 30tons per month.

- *Introduce and replace the current machinery for new machineries*

The currently used cutting machines will be replaced with a new machinery. It increases the production rate and could increase the supply for a very high demand. Also, the life span is much higher, and the power required is less compared to the old machinery.

Table 1: Production rates

Old cutting machine	New cutting machine
Can cut 3500 husks per day	Can cut 5000 husks per day

- *Alternative drying methods suitable for any weather condition*

The coco chips must be dry up to minimum of 80% according to industry standards. The current way is drying under sunset in a large area for about 8 hours.

The current method is replaced with a new drying machine which can be used under any weather condition and reliable with the cost as it can produce a high supply demand in a minimize time.

- *Plan the production process and schedule the production line processes*

The current production process from storing the husks to packaging the products doesn't align with a step-by-step process. This means the outputs from each machine is not send to the next machines accurately. Its reasons for a much higher wastage in each process. Therefore, a more utilized and optimized layout is introduced to continue from one step to another.

When the process is made in a proper manner more space can be obtained which will be helpful on drying coco peals and categorizing peals according to the market size values.

The schedule time is optimized to minimize time for a process and improve productivity. This can reduce the labour cost as well. Schedule time on drying coco peals, cutting husks, sorting out the quality raw materials for the production can be done by the methodologies used above.

The process optimization is performed by using plant simulation software and the results are validated to get the optimum production rate with minimized cost.

2.3 Simulation and analysis

2.3.1 Analysis

Analysis for coco chips production

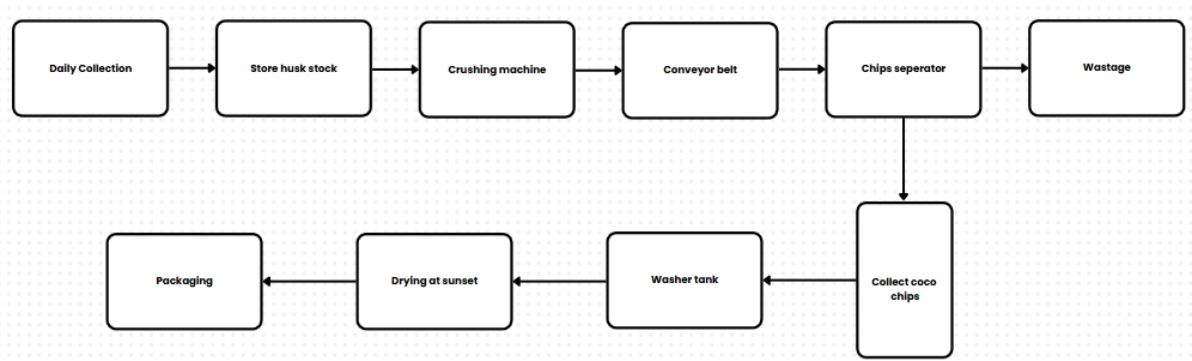


Figure 4: Process flow for coco chips

First, from the given data we considered 5000husks which is approximately 2tons are collected which aligns with the market demand.

Therefore, by using the current machines the husks can be cut from nearly 4hrs.

The husking rate of currently used machines vary in a range of 2500 to 3000. Here, two labors are used for inset husks to the machine. Another labor collects the coco chips and load to the conveyor belt.

The chips are sent through the conveyor belt and send to the chips separator and obtain the industrial standards chip sizes. It will take approximately 1.5hours. It depends on the chip sizes and moisture content.

But due to the currently used cutting machines from the input husks 40% wastage occurs.

There are two types of coco chips used in industry. They are washed and unwashed. But washed coco chip products has a high demand. To obtain the washed product the coco chips are kept in a huge water tank for 12 hours to remove the acidic on them.

The chips are dried on sunset for minimum 8hours till the moisture content is reduced to 20% and it can be measured from the moisture tester.

The chips are stored and packaged for distribution which will take about 1hour.

Analysis for coco-fibre

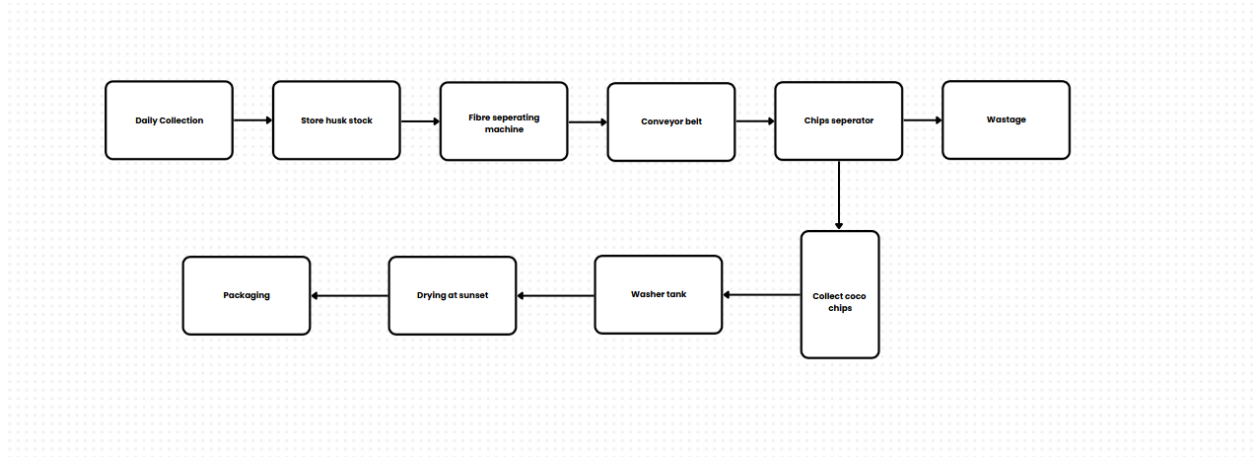


Figure 5: Process flow for coco-fibre

A fibre separating machine is used to obtain coco-fibre which separates 40% of fibre and rest of the process is same used for coco peat.

2.3.2 Simulation

The Siemens Technomatrix Plant Simulation software is used to the simulation and analysis, which is a specialized tool built for manufacturing process modeling and optimization. A comprehensive digital model of the production line with accurately replicating the flow of materials, machine operations, and worker interactions are done using the software. It visually showcases areas of inefficiency within the production process. The simulation was done by the

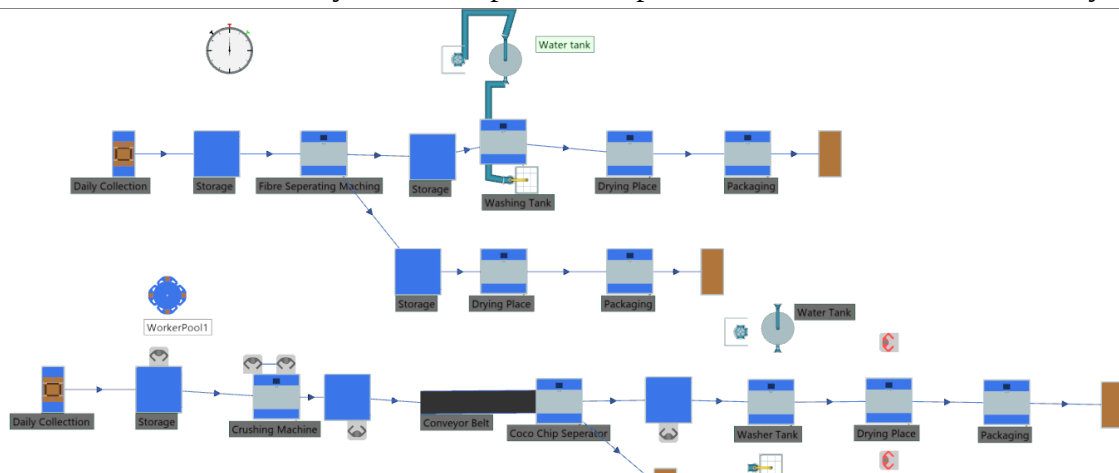


Figure 6: Simulation of current production line

collected data on each process and current production line. The base calculations are done using data on the current machinery data and capacity of them. The time values are obtained by the coworkers based on the ongoing production line.

The proposed solution is done by implementing new workstations, and repositioning existing workstations. The optimized simulation improved the overall production efficiency and performance while reducing labor cost currently required.

In the solution simulation the old crushing machine and drying method is replaced with new machinery. It optimizes the old production line and improve time efficiency and could able to improve supply over high demand.

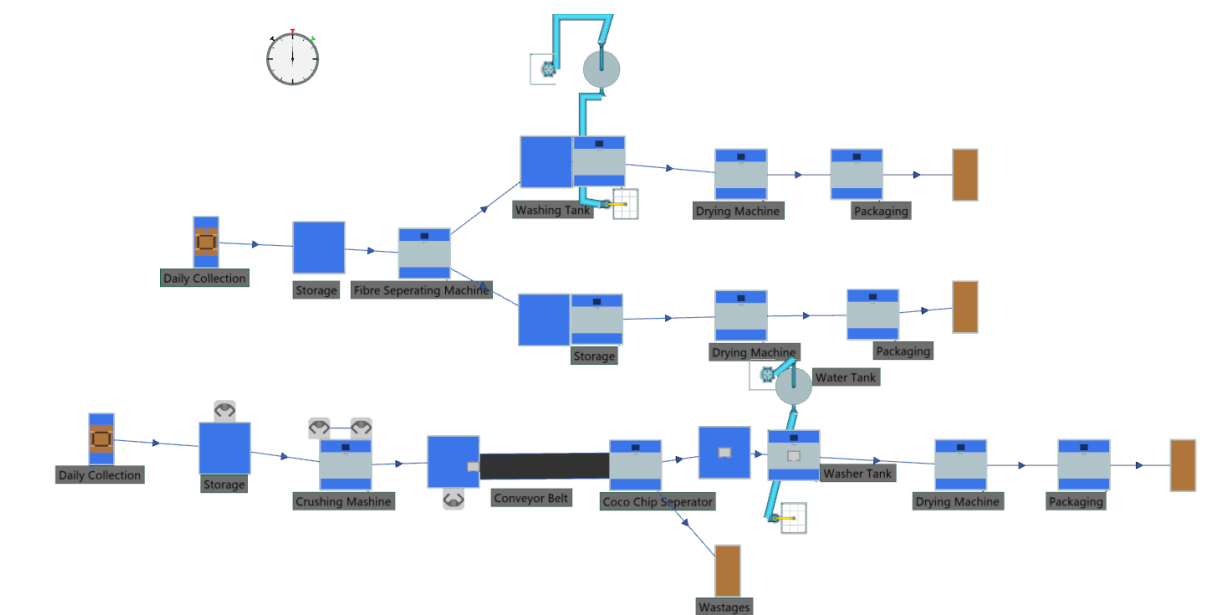


Figure 7: Simulation of optimized production line

Simulation Video link

<https://dms.uom.lk/s/FZmsCAbMssPbY2C>

3 Findings

3.1.1 Modifications

Table 2: Change in performance

Old Production line	Optimized production line
Husk cutting capacity – 2000-3000 hours per day	Husk cutting capacity – 3000-5000 hours per day
Time duration for husk cutting – 4hours	Around 2hours.
High wastage (40%)	Less wastage (<20%)
Coco chips can only dry under sunset	Drying machine is introduced
Minimum 8hours required to dry and maintain the moisture content	Can be dry soon after the chips are washed
Current number of labors – 15 (Labor cost is high)	Maximum of 10 labors (Less labor costs)

- Replacement of a new cutting machine which can insert husks from two sides which increases the production rate.

Coconut Husk Chips Machine – Double Side



Figure 8: Husk crushing machine

- Replacement of a drying machine (Coco peat dryer) without using sunset as the energy source. This might cost for electricity, but it improves the efficiency of the production line.
 - Machine capacity – 300 – 500kg/h
 - Motor power – 2HP/1.5kW
 - Purposes – Drying of paddy husk, coco peat

3.1.2 Results

Current Production Rate

Waiting Time					
Object	Portion	Count	Sum	Mean Value	Standard Deviation
Source	100.00%	2	1:01:20:01.3000	12:40:00.6500	3:46:17.3694
Station1	68.42%	2	17:20:01.3000	8:40:00.6500	3:46:17.3694
Station2	84.21%	1	21:20:01.3000	21:20:01.3000	0.0000
Station3	100.00%	3	1:01:19:58.7000	8:26:39.5667	3:35:42.6705
Station5	52.63%	2	13:20:01.3000	6:40:00.6500	3:17:58.4747
Buffer	100.00%	2	1:01:20:01.3000	12:40:00.6500	3:46:17.3694
Buffer1	100.00%	2	1:01:20:01.3000	12:40:00.6500	11:47:05.4876
Conveyor	0.00%	0	0.0000	0.0000	0.0000
Station4	68.42%	2	17:20:01.3000	8:40:00.6500	10:50:33.2136
Station6	96.05%	1	1:00:20:01.3000	1:00:20:01.3000	0.0000
Drain2	100.00%	1	1:01:20:01.3000	1:01:20:01.3000	0.0000
Station7	52.63%	2	13:20:01.3000	6:40:00.6500	3:46:17.3694
Station8	68.42%	2	17:20:01.3000	8:40:00.6500	9:25:42.0447
Buffer3	100.00%	2	1:01:20:01.3000	12:40:00.6500	12:15:24.3824
Buffer4	100.00%	2	1:01:20:01.3000	12:40:00.6500	15:05:06.7200
Station9	96.05%	2	1:00:20:01.3000	12:10:00.6500	3:03:51.7850
Drain	100.00%	2	1:01:20:01.3000	12:40:00.6500	2:21:26.2006
Station10	68.42%	2	17:20:01.3000	8:40:00.6500	10:22:14.3188
Station11	99.99%	2	1:01:19:51.3000	12:39:55.6500	16:01:46.0652
Drain1	100.00%	2	1:01:20:01.3000	12:40:00.6500	16:01:53.1362
Buffer2	100.00%	3	1:01:20:01.3000	8:26:40.4333	3:54:23.1974
Source1	100.00%	1	1:01:20:01.3000	1:01:20:01.3000	0.0000
Buffer5	100.00%	1	1:01:20:01.3000	1:01:20:01.3000	0.0000

Figure 9:Simulation results of current production

Optimized production process

Working Time					
Object	Portion	Count	Sum	Mean Value	Standard Deviation
Source	0.00%	0	0.0000	0.0000	0.0000
Station2	68.00%	3	17:00:00.0000	5:40:00.0000	34:38.4610
Station3	0.01%	3	6.6600	2.2200	0.0000
Station5	98.62%	3	1:00:39:14.2670	8:13:04.7557	6:33:02.2950
Buffer	0.00%	0	0.0000	0.0000	0.0000
Buffer1	0.00%	0	0.0000	0.0000	0.0000
Station	0.01%	3	7.5000	2.5000	0.0000
Conveyor	100.00%	1	1:01:00:00.0000	1:01:00:00.0000	0.0000
Station4	0.01%	2	5.0000	2.5000	0.0000
Station6	6.57%	2	1:38:32.7968	49:16.3984	15:10.1901
Drain2	0.00%	1	0.0000	0.0000	0.0000
Station7	48.00%	1	12:00:00.0000	12:00:00.0000	0.0000
Station8	36.00%	1	9:00:00.0000	9:00:00.0000	0.0000
Buffer3	0.00%	0	0.0000	0.0000	0.0000
Buffer4	0.00%	0	0.0000	0.0000	0.0000
Station9	0.00%	1	0.0000	0.0000	0.0000
Station10	0.00%	1	2.5000	2.5000	0.0000
Station11	4.00%	1	1:00:00.0000	1:00:00.0000	0.0000
Drain1	0.00%	1	0.0000	0.0000	0.0000
Buffer2	0.00%	0	0.0000	0.0000	0.0000
Buffer5	0.00%	0	0.0000	0.0000	0.0000
Source1	0.00%	0	0.0000	0.0000	0.0000
Station1	0.00%	0	0.0000	0.0000	0.0000

Figure 10:Simulation results of optimized production

Material Flow Properties							
Object	Number of Entries	Number of Exits	Minimum Contents	Maximum Contents	Relative Empty	Relative Full	Relative Occupation without Interruptions
Source	2	2	0	1	100.00%	-	0.00%
Station1	2	2	0	1	68.42%	-	31.58%
Station2	2	2	0	1	84.21%	-	15.79%
Station3	2	2	0	1	100.00%	-	0.00%
Station5	1	1	0	1	52.63%	-	47.37%
Buffer	2	2	0	1	100.00%	0.00%	0.00%
Buffer1	1	1	0	1	100.00%	0.00%	0.00%
Conveyor	2	2	0	1	97.37%	-	0.25%
Station4	1	1	0	1	68.42%	-	31.58%
Station6	1	1	0	1	96.05%	-	3.95%
Drain2	1	1	0	1	100.00%	-	0.00%
Station7	1	1	0	1	52.63%	-	47.37%
Station8	1	1	0	1	68.42%	-	31.58%
Buffer3	1	1	0	1	100.00%	0.00%	0.00%
Buffer4	1	1	0	1	100.00%	0.00%	0.00%
Station9	1	1	0	1	96.05%	-	3.95%
Drain	1	1	0	1	100.00%	-	0.00%
Station10	1	1	0	1	68.42%	-	31.58%
Station11	1	1	0	1	99.99%	-	0.01%
Drain1	1	1	0	1	100.00%	-	0.00%
Buffer2	2	2	0	1	100.00%	0.00%	0.00%
Source1	2	2	0	1	100.00%	-	0.00%
Buffer5	2	2	0	1	92.11%	0.00%	0.99%
Drain3	1	1	0	1	100.00%	-	0.00%

Material Flow Properties							
Object	Number of Entries	Number of Exits	Minimum Contents	Maximum Contents	Relative Empty	Relative Full	Relative Occupation without Interruptions
Source	3	3	0	1	100.00%	-	0.00%
Station2	3	2	0	1	32.00%	-	68.00%
Station3	3	3	0	1	99.97%	-	0.03%
Station5	3	2	0	1	1.38%	-	98.62%
Buffer	3	3	0	1	99.97%	0.00%	0.00%
Buffer1	3	3	0	1	75.99%	0.00%	0.00%
Station	3	3	0	1	99.94%	-	0.06%
Conveyor	3	3	0	1	96.00%	-	0.38%
Station4	2	2	0	1	99.95%	-	0.05%
Station6	2	1	0	1	93.38%	-	6.62%
Drain2	1	1	0	1	100.00%	-	0.00%
Station7	1	1	0	1	52.00%	-	48.00%
Station8	1	1	0	1	64.00%	-	36.00%
Buffer3	1	1	0	1	100.00%	0.00%	0.00%
Buffer4	1	1	0	1	100.00%	0.00%	0.00%
Station9	1	0	0	1	100.00%	-	0.00%
Station10	1	1	0	1	100.00%	-	0.00%
Station11	1	1	0	1	96.00%	-	4.00%
Drain1	1	1	0	1	100.00%	-	0.00%
Buffer2	3	3	0	1	100.00%	0.00%	0.00%
Buffer5	3	3	0	1	100.00%	0.00%	0.00%
Source1	3	3	0	1	100.00%	-	0.00%
Station1	0	0	0	0	100.00%	-	0.00%
Drain	0	0	0	0	100.00%	-	0.00%

Working Time

Object	Portion	Count	Sum	Mean Value	Standard Deviation
Source	0.00%	0	0.0000	0.0000	0.0000
Station1	31.58%	2	8:00:00.0000	4:00:00.0000	0.0000
Station2	15.79%	2	4:00:00.0000	2:00:00.0000	0.0000
Station3	0.00%	2	2.6000	1.3000	0.0000
Station5	47.37%	1	12:00:00.0000	12:00:00.0000	0.0000
Buffer	0.00%	0	0.0000	0.0000	0.0000
Buffer1	0.00%	0	0.0000	0.0000	0.0000
Conveyor	100.00%	1	1:01:20:01.3000	1:01:20:01.3000	0.0000
Station4	31.58%	1	8:00:00.0000	8:00:00.0000	0.0000
Station6	3.95%	1	1:00:00.0000	1:00:00.0000	0.0000
Drain2	0.00%	1	0.0000	0.0000	0.0000
Station7	47.37%	1	12:00:00.0000	12:00:00.0000	0.0000
Station8	31.58%	1	8:00:00.0000	8:00:00.0000	0.0000
Buffer3	0.00%	0	0.0000	0.0000	0.0000
Buffer4	0.00%	0	0.0000	0.0000	0.0000
Station9	3.95%	1	1:00:00.0000	1:00:00.0000	0.0000
Drain	0.00%	1	0.0000	0.0000	0.0000
Station10	31.58%	1	8:00:00.0000	8:00:00.0000	0.0000
Station11	0.01%	1	10.0000	10.0000	0.0000
Drain1	0.00%	1	0.0000	0.0000	0.0000
Buffer2	0.00%	0	0.0000	0.0000	0.0000
Source1	0.00%	0	0.0000	0.0000	0.0000
Buffer5	0.00%	0	0.0000	0.0000	0.0000

Waiting Time

Object	Portion	Count	Sum	Mean Value	Standard Deviation
Source	100.00%	3	1:01:00:00.0000	8:20:00.0000	2:53:12.3048
Station2	32.00%	2	8:00:00.0000	4:00:00.0000	0.0000
Station3	99.97%	4	1:00:59:34.7209	6:14:53.6802	4:40:36.4054
Station5	1.38%	1	20:45.7330	20:45.7330	0.0000
Buffer	100.00%	3	1:01:00:00.0000	8:20:00.0000	2:53:12.3048
Buffer1	100.00%	4	1:01:00:00.0000	6:15:00.0000	4:40:31.2669
Station	99.98%	4	1:00:59:37.8894	6:14:54.4723	4:47:16.0489
Conveyor	0.00%	0	0.0000	0.0000	0.0000
Station4	99.99%	3	1:00:59:55.0000	8:19:58.3333	6:39:26.6294
Station6	93.43%	2	23:21:27.2032	11:40:43.6016	57:35.7745
Drain2	100.00%	2	1:01:00:00.0000	12:30:00.0000	1:12:45.9646
Station7	52.00%	2	13:00:00.0000	6:30:00.0000	42:25.5844
Station8	64.00%	1	16:00:00.0000	16:00:00.0000	0.0000
Buffer3	100.00%	2	1:01:00:00.0000	12:30:00.0000	9:11:32.5974
Buffer4	100.00%	2	1:01:00:00.0000	12:30:00.0000	4:56:59.0909
Station9	100.00%	1	1:01:00:00.0000	1:01:00:00.0000	0.0000
Station10	100.00%	2	1:00:59:57.5000	12:29:58.7500	7:46:43.1963
Station11	96.00%	2	1:00:00:00.0000	12:00:00.0000	8:29:10.5485
Drain1	100.00%	2	1:01:00:00.0000	12:30:00.0000	9:11:36.1329
Buffer2	100.00%	4	1:01:00:00.0000	6:15:00.0000	4:47:13.9339
Buffer5	100.00%	3	1:01:00:00.0000	8:20:00.0000	2:53:12.3048
Source1	100.00%	3	1:01:00:00.0000	8:20:00.0000	2:53:12.3048
Station1	100.00%	1	1:01:00:00.0000	1:01:00:00.0000	0.0000
Drain	100.00%	1	1:01:00:00.0000	1:01:00:00.0000	0.0000

The above are the simulation results obtained which shows the optimized production line.

3.2 Discussion and recommendations

- Upgrade machinery

The organization should continue investing in modern machinery, such as the husk cutting and drying machines, to improve production capacity and operational efficiency. The cost savings from reduced wastage and labor, coupled with increased output, should justify the investment in the long term.

- Enhance Quality Control

A more scientific approach to assessing the quality of coconut husks, rather than relying on visual identification, will reduce wastage further and ensure that only high-quality raw materials enter the production process. Implementing automated quality testing procedures, such as moisture and fiber content tests, can help ensure consistent product standards.

- Continuous Monitoring and Simulation

Regularly using simulation tools like Siemens Technomatrix Plant Simulation should be part of the facility's strategy. By continuously monitoring the process, the facility can spot any problems or slowdowns as production increases or market demands change, helping the facility quickly adapt to those changes.

- Implement Advanced Demand Forecasting Systems

The demand forecasting system, based on historical data, should be implemented to manage raw material supply better, reduce transportation costs, and optimize inventory levels. This will help balance the fluctuations in supply and demand, particularly during periods of high demand or raw material shortages.

The below charts show the demand for husks on 2021/22/23 years. Also, on January, February and March months the production process is stopped due to the rainy conditions.

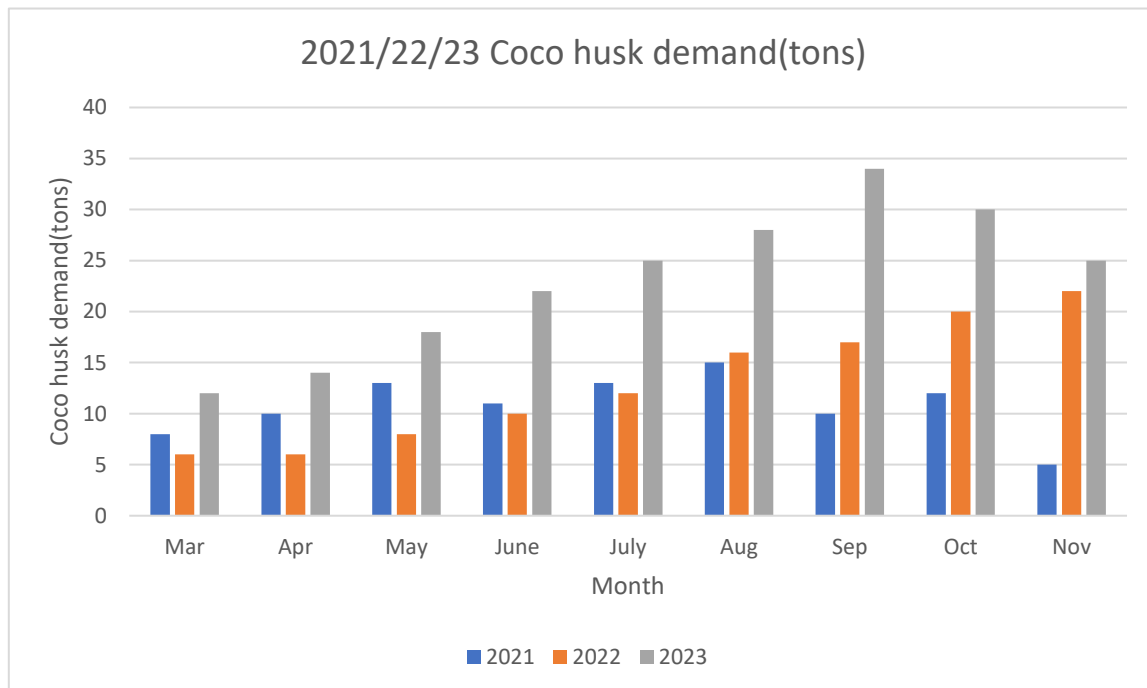


Figure 11:Coco husk demands of 2021/22/23

A seasonal index is obtained for each month by dividing the average demand for month by the total average demand which is 15.62963tons. The forecast demand is calculated by the total average demand by each month's seasonal index.

Table 3:Forecasted demand of 2021/22/23

Month	Coco husk demand (tons)			Average Demand	Seasonal Index	Forcasted Demand
	2021	2022	2023			
Jan						
Feb						
Mar	8	6	12	8.666666667	0.55450237	8.666666667
Apr	10	6	14	10	0.639810427	10
May	13	8	18	13	0.831753555	13
June	11	10	22	14.33333333	0.917061611	14.33333333
July	13	12	25	16.66666667	1.066350711	16.66666667
Aug	15	16	28	19.66666667	1.258293839	19.66666667
Sep	10	17	34	20.33333333	1.300947867	20.33333333
Oct	12	20	30	20.66666667	1.322274882	20.66666667
Nov	5	22	25	17.33333333	1.109004739	17.33333333
Dec						

The seasonal factors are obtained for each month and average is considered. It is used to find the demand or forecasted demand values that can be in year 2024. The below graph shows the forecasted demand for each month.

Table 4:Forecasted demand for 2024

Month	Saesonal Factor			Average Seasonal Factor	Forecasted Demand	Rounded Demand
	2021	2022	2023			
Jan						
Feb						
Mar	0.5118	0.3839	0.7678	0.5545	8.666666667	9
Apr	0.6398	0.3839	0.8957	0.6398	10	10
May	0.8318	0.5118	1.1517	0.8318	13	13
June	0.7038		1.4076	1.0557	16.5	17
July	0.8318	0.7678	1.5995	1.0664	16.66666667	17
Aug	0.9597	1.0237	1.7915	1.2583	19.66666667	20
Sep	0.6398	1.0877	2.1754	1.3009	20.33333333	21
Oct	0.7678	1.2796	1.9194	1.3223	20.66666667	21
Nov		1.4076	1.5995	1.5036	23.5	24
Dec						

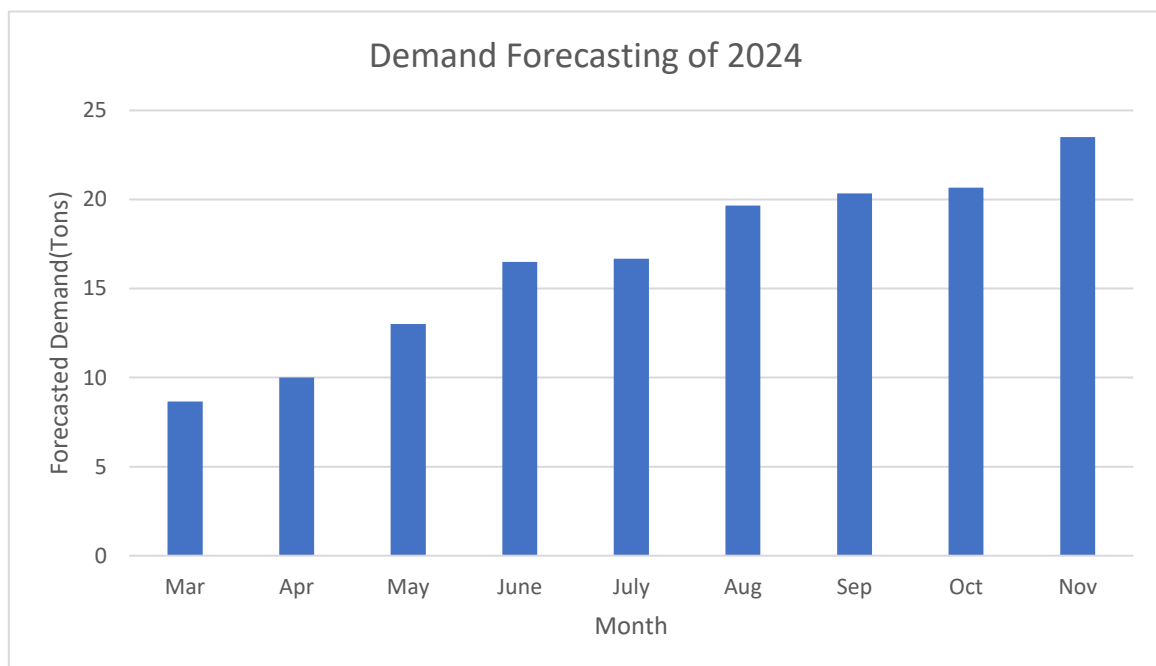


Figure 12:Graph of Demand forecasting of 2024

The obtained forecasting graph data can be used to schedule and plan the future production and plan the husk collecting with less wastage as well.

- Flexible Production Layout and Process Flow Optimization

The organization should continue optimizing its production layout to streamline the workflow between workstations. The improvements in space utilization, as seen in the drying and packaging process, should be expanded to other parts of the production line, ensuring smooth transitions between stages and minimizing material handling times.

4 Timeline

	Week													
Activity	1	2	3	4	5	6	7	8	9	10	11	12	13	14
Form a group														
Choose a company to visit														
Factory visit														
Identification of problems														
Prepare a project proposal														
Analysis of problems														
Discuss the possible solutions														
Prepare an interim report														
Prepare a final report														
Presentation														

Figure 13: Timeline

5 Estimated budget

Table 5: Estimated budget

Raw Material	Cost per kg (or batch)	Quantity Used	Quantity Used (kg)	Daily Cost	Monthly Cost
Coconut husk	10.77	5000	1625	17500	420000
Number of Workers	Daily Wage	Hours Worked	Total Labor Cost		
10	1500	192	360000		
Monthly Energy Cost	19000				
Products	Units of Product Produced (kg)	Cost per Unit of Product			
Coco Chips	23400	40			
Total cost per kg	34.14				
Price per kg	40				
Profit per kg	5.86				
Total Monthly Cost	799000				
Total Monthly Price	936000				
Total Monthly Profit	137000				

6 Limitations and future work

- The durability of coconut husks is very less to use as one of our products because they are infected from insects very quickly and even, they are kept in the stagnate tank for considerable amount of time it can cause other environmental issues.

Therefore, a considerable amount of wastage occurs as the exact raw material quantity doesn't coincide with the market demand. Also, when making the products coco chips and coco peels a wastage occurs as well. Therefore, again the same issue occurs and drive into a wastage of raw material.

- There is a possibility of ignition of coco chips as they consist of a larger surface area when laid on sunset. If the ignition temperature equals with the current temperature of the surrounding, there can be a possibility of fire. Therefore, there is a considerable risk as well.
- Adjusting with varying weather conditions is an issue. Therefore, it is hard to plan the production layout for an optimum condition.

Future Improvements

- Automate the production line that can be used under any circumstance or condition.
- Reduce the wastage by testing more samples and make a data analysis system to manage the supply and demand.

7 References

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